

NEDA FALLAH

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EDUCATION

Sharif University of Technology, Tehran, Iran

Sep. 2019 - July 2024

B.Sc. in Applied Mathematics,

- Minor: Computer Science
- GPA: 3.3/4

RESEARCH INTERESTS

- Natural Language Processing
- Multimodal Learning
- Generative AI
- Computer Vision
- Information Retrieval
- Machine Learning & Deep Learning
- Data Science
- Speech Processing

RESEARCH EXPERIENCE

**Emo3D: Translating Emotion Descriptions to 3D Facial Expressions,
Benchmark Dataset and Evaluation Metrics**

June 2023 - April 2024

Research Assistant | Supervisor: Dr. Ehsaneddin Asgari, Sharif University

- Contributed to the development of **Emo3D**, a benchmark dataset for text-guided 3D facial expression generation. Preprint ([Link](#))
- Developed a benchmark Dataset and evaluation metrics.
- Baseline and CLIP-based models were evaluated on this dataset

Important News Detection: Development of a Persian Large Language Model

Aug. 2023 - April 2024

Research Assistant | Supervisor: Prof. Mahdi Jafari Siavoshani, INL Lab, Sharif University

- Designed and implemented web scrapers for Persian news.
- Fine-tuned open-source LLMs on a self-crawled dataset of Persian news articles.
- Detection of important news using this model.

Exploring Attention Mechanisms and Transformer Models

July 2022 - Oct. 2022

Bachelor's Thesis | Supervisor: Prof. Mojtaba Tefagh, Sharif University

- Researched the application of attention mechanisms and transformer models in NLP and computer vision.
- Presented at a Deep Learning Webinar at Sharif University.

PREPRINTS

**Emo3D: Translating Emotion Descriptions to 3D Facial Expressions,
Benchmark Dataset and Evaluation Metrics**

[Link to preprint](#)

TEACHING EXPERIENCE

- **Teaching Assistant**, Sharif University: *Feb. 2021 - Feb. 2024*
Machine Learning (2 times), Linear Algebra (4 times), Probability and Statistics, Discrete Mathematics, Automata Theory
- **Instructor**, Isfahan Mathematics House: *Winter/Summer 2021*
Mathematical Modelling Class (Recommender Systems)

HONORS

- Ranked Top 0.25% in the Iranian University Entrance Exam (Konkour)
- Awarded full scholarship at Sharif University for Bachelor's degree, Summer 2019
- Qualified in the Iranian National Olympiad in Informatics (Iran IOI), Spring 2017
- Honored in the International Mathematics Tournament of Towns, Winter 2016
- Ranked 2nd in Exceptional Talent School Entrance, Spring 2013

RELEVANT COUREWORK

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|--|------------------------------|
| • Natural Language Processing(Graduate Course) | • Analysis of Algorithms |
| • Machine Learning | • Probability and Statistics |
| • Statistical Learning | • Linear Algebra |
| • Data Structures | • Speech Processing |

PROJECTS

- Persian CLIP (Joint Latent Space Learning) - Natural Language Processing Course Homework, Python
- Sentiment Analysis-Question Answering - Natural Language Processing Course Homework, Python
- Breast Cancer Survival Prediction - Machine Learning Course Project, Python
- Statistical Learning Methods Applied to Iran's Health Dataset - Regression Analysis Course Project, R

WORK EXPERIENCE

Maani Studio, Creative Developer *Mar 2024-Present*

- Enhance animations and films with AI tools to reduce costs and speed production.
- Developed Blender add-ons using Python.
- Researched Computer Vision and Multimodal (Text-Image) Integration.

RZ Company, AI Developer *Mar 2024-Present*

- Developed and integrated facial recognition and AI avatar features.
- Optimized an LLM-based chatbot and created supporting APIs.
- Conducted quality assurance for enhanced website performance.

SKILLS

- **Programming Languages:** Python, Java, R, MATLAB, HTML/CSS
- **Libraries:** PyTorch, TensorFlow, OpenCV, Matplotlib, Keras, Transformers, Scikit-learn, Pandas
- **Tools:** LaTeX, Git/Github, Excel, Photoshop, AI APIs
- **Databases:** PostgreSQL, MongoDB, SQL
- **Frameworks:** Django, Node.js
- **Soft Skills:** Critical Thinking, Communication, Adaptability, Teamwork, Creativity

LANGUAGES

- **English** (Proficient, TOEFL iBT: 113, Reading 30, Listening 30, Speaking 27, Writing 26)
- **Farsi** (Native)
- **German** (Elementary)
- **Italian** (Beginner)
- **Arabic** (Beginner)

VOLUNTEERING

- **Rasta**, Sharif University, Mentor: Workshops on Image Processing and Recommender Systems
- **Basketball Team**, Sharif University, Member
- **English Free Discussions**, Sharif University, Organizer
- **Isfahan Mathematics Karsough**, Mentor: Workshops on Fractals and Logic Puzzles
- **Isfahan Mathematics House**, Instructor: Recommender Systems Class
- **Isfahan Mathematics House**, Member of Combinatorics Research Group: Worked on the math behind the card game "Spot-it"
- **Hamband Scientific Society**, Sharif University, Graphic Designer

REFERENCES

Dr. Bahar Farahani,
Assistant Professor, Shahid Beheshti University

[Google Scholar Profile](#)

Dr. Mojtaba Tefagh,
Assistant Professor, Sharif University

[Google Scholar Profile](#)

Dr. Ehsaneddin Asgari,
Research Scientist, Qatar Computing Research Institute

[Google Scholar Profile](#)

Dr. Mahdi Jafari Siavoshani,
Assistant Professor, Sharif University

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